

WESTERN MICHIGAN, MI, USA

WMO: 725394

Lat: 42.746N

Lon: 86.097W

Elev: 689

StdP: 14.33

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 04839

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.6	9.8	-1.9	5.1	7.3	1.8	6.2	11.0	28.7	34.3	26.7	33.7	7.3	270	0.542

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.6	89.1	73.4	86.3	72.0	83.5	70.8	76.2	85.3	74.5	82.8	73.0	80.7	11.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.1	126.1	81.6	72.0	121.3	80.0	70.3	114.4	78.0	40.0	85.5	38.4	83.4	36.9	80.5	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.6	22.0	19.1	-2.6	92.9	5.8	3.0	-6.8	95.0	-10.1	96.8	-13.4	98.4	-17.6	100.6
			-3.3	78.8	5.6	2.1	-7.3	80.4	-10.6	81.6	-13.7	82.8	-17.8	84.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.8	26.3	27.8	36.3	47.9	58.7	68.0	72.1	70.2	63.9	52.4	41.4	31.8
	DBStd	18.22	9.89	9.59	10.91	9.09	8.77	6.92	5.59	5.08	7.32	8.30	8.43	8.68
	HDD50	2894	736	622	450	145	21	0	0	0	1	69	281	569
	HDD65	6201	1200	1041	892	519	238	46	7	13	106	401	707	1031
	CDD50	2838	0	1	25	81	289	541	685	627	419	143	24	3
	CDD65	671	0	0	2	5	41	137	227	175	74	10	0	0
	CDH74	5582	0	0	15	47	375	1139	2029	1308	592	77	0	0
	CDH80	1535	0	0	2	5	99	337	627	303	152	10	0	0
Wind	WSAvg	8.6	10.2	9.8	9.5	9.7	8.6	7.5	6.6	6.1	6.7	8.4	9.7	10.1
Precipitation	PrecAvg	35.4	2.1	1.8	2.1	3.4	4.1	3.3	2.6	3.2	3.3	3.8	2.8	2.8
	PrecMax	52.0	4.1	3.4	4.0	7.2	9.3	8.9	5.8	6.8	10.9	8.9	8.1	7.6
	PrecMin	27.0	0.7	0.7	0.7	0.7	1.3	0.9	0.4	1.0	0.4	0.6	0.6	1.3
	PrecStd	6.7	1.0	0.7	1.1	1.7	2.0	2.3	1.3	1.7	2.4	2.2	1.7	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.5	59.9	74.9	80.1	88.0	91.1	93.4	90.2	89.1	81.5	68.8	59.5
		MCWB	51.5	54.4	62.9	63.1	70.4	73.8	76.1	74.5	70.6	68.9	57.2	55.5
	2%	DB	48.0	52.0	65.9	73.3	82.4	87.7	89.6	86.3	84.1	75.2	63.0	53.7
		MCWB	44.8	46.7	57.8	59.2	67.5	71.9	74.7	72.8	71.0	64.9	55.6	51.0
	5%	DB	43.2	45.0	59.0	68.5	78.2	83.8	86.3	83.2	80.8	70.8	58.7	47.9
		MCWB	39.6	40.3	51.6	55.8	65.2	70.1	72.7	71.2	68.5	63.0	51.9	44.5
	10%	DB	38.9	40.0	52.4	63.4	73.5	80.9	83.3	80.9	76.9	66.2	54.6	43.3
		MCWB	36.5	36.6	45.7	53.0	62.4	69.2	71.4	70.1	66.9	59.6	49.1	39.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.3	54.7	63.4	66.1	72.8	77.0	79.0	77.4	75.4	70.9	60.1	56.6
		MCDB	54.9	57.3	74.0	76.2	83.9	86.9	88.7	86.2	83.8	79.1	64.6	59.1
	2%	WB	45.5	46.3	58.6	61.2	69.5	74.3	76.9	75.1	72.9	67.2	56.7	50.6
		MCDB	47.2	51.4	65.0	70.2	79.2	84.0	86.4	82.8	80.2	72.5	62.1	53.3
	5%	WB	40.0	40.7	52.8	57.5	67.2	72.3	75.0	73.6	70.7	63.9	53.0	44.7
		MCDB	42.6	44.4	57.9	66.0	75.5	81.1	83.6	80.4	77.5	68.7	57.7	47.2
	10%	WB	36.7	37.1	46.0	53.8	64.6	70.4	73.2	72.0	68.6	60.1	49.5	40.1
		MCDB	39.2	40.0	51.5	62.1	72.0	78.4	81.1	78.6	75.3	66.2	54.6	43.3
Mean Daily Temperature Range	5% DB	MDBR	11.1	12.6	15.3	18.4	18.8	18.3	18.6	18.4	19.5	16.7	13.0	10.1
		MCDBR	15.5	18.0	22.4	26.0	23.1	21.5	21.2	21.2	22.9	22.7	19.1	15.6
		MCWBR	13.4	14.1	15.4	15.3	12.8	10.4	10.0	10.6	11.5	13.9	13.9	14.0
	5% WB	MCDBR	14.2	17.2	20.8	22.3	20.7	18.4	18.3	16.7	18.6	19.8	17.3	15.2
		MCWBR	13.1	14.4	15.7	15.1	12.7	10.5	10.0	9.7	10.8	13.9	14.1	14.4
Clear-Sky Solar Irradiance	taub	0.301	0.339	0.344	0.379	0.411	0.428	0.425	0.424	0.393	0.353	0.321	0.301	
	taud	2.403	2.269	2.365	2.297	2.268	2.272	2.281	2.283	2.341	2.442	2.477	2.445	
	Ebn at Noon	266	270	285	281	274	269	268	265	265	263	257	255	
	Edh at Noon	25	33	34	40	42	42	41	40	35	28	23	22	
All-Sky Solar Radiation	RadAvg	390	647	1124	1502	1748	1974	2009	1713	1366	851	500	322	
	RadStd	42	88	104	135	161	129	89	98	120	99	75	45	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (62 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page